Physical loading model

Loading and event decrements

$$X(t) = X_0\Phi_s(t) + \rho\int_0^t \Phi_s(u)\,du - G_s(t)$$

Physical intensity

$$\lambda_{\text{phys}} = \Psi(X(t^-)) + H_{\text{phys}}(t)$$

Independent-background diagnostic

Background and event memory

$$B(t), \quad G_s(t) = \sum_{t_i < t} g(Q_i)\Phi_s(t - t_i)$$

Diagnostic intensity

$$\lambda_\eta = B(t)e^{-\eta G_s(t)} + nH_e(t)$$

Shared-memory restriction: $h_e = -\Phi_s'$

The background B is independent of the physical loading construction.